



Lounge Eligibility Lookup

Grouping	Example Destinations	Tier 1 %	Tier 2 %	Tier 3 %	Notes
LONG - Peak Business (PB)	LAX, JFK, BOS, SFO, ORD (North America); NRT, HKG (Asia)	1.48%	20.42%	78.10%	Morning departures (06:00-10:00). High proportion of business travelers and premium cabin bookings.
LONG - Off-Peak/Leisure (OP)	LAX, MIA, SAN, SEA (North America); DXB, BKK (Asia/Middle East)	1.47%	20.49%	78.04%	Midday/afternoon departures (10:00-19:00). Mix of business and leisure travelers.
LONG - Evening/Sleeper (ES)	SIN, HKG, BOM, DEL (Asia); JFK, IAD (North America overnight)	1.58%	20.39%	78.03%	Evening/night departures (19:00-06:00). Premium travelers preferring overnight flights and lie-flat seats.
SHORT - Peak Business (PB)	FRA, CDG, AMS, ZRH, MAD (Europe business hubs)	1.55%	20.38%	78.07%	Morning departures (06:00-10:00). Dominated by business travelers on day trips.
SHORT - Off-Peak/Leisure (OP)	BCN, FCO, ATH, LIS, DUB (Europe leisure destinations)	1.71%	20.32%	77.96%	Midday/afternoon departures (10:00-19:00). Higher leisure mix, still substantial loyalty members.
SHORT - Evening/Sleeper (ES)	IST, ATH, MXP, VCE (Europe evening routes)	1.51%	20.42%	78.07%	Evening departures (19:00-06:00). Mix of business travelers extending meetings and leisure travelers.



Justification:

Prompt	Your Response
1. How did you group the flights in your table?	I grouped flights using a two-dimensional classification system, which are Dimension 1: Route Type (HAUL) LONG haul: Intercontinental flights (typically >6 hours) - Examples: LAX, JFK, HKG, SIN SHORT haul: European and regional flights (typically <6 hours) - Examples: FRA, CDG, BCN, IST. Dimension 2: Time of Day (TIME_CATEGORY) Peak Business (PB): 06:00-10:00 AM departures Off-Peak/Leisure (OP): 10:00-19:00 (7:00 PM) departures Evening/Sleeper (ES): 19:00-06:00 (overnight) departures This created 6 distinct flight groups: LONG - Peak Business LONG - Off-Peak/Leisure LONG - Evening/Sleeper SHORT - Peak Business SHORT - Off-Peak/Leisure SHORT - Evening/Sleeper
2. Why did you choose that grouping method?	This grouping method was chosen for several strategic reasons: A. Business Behavior Patterns Morning flights (Peak) attract business travelers who need same-day returns or early meetings - these passengers typically have higher loyalty status. Midday/afternoon flights (Off-Peak) have a mixed customer base of both business and leisure travelers Evening flights (Sleeper) appeal to premium travelers wanting overnight comfort and maximizing time at destination B. Route Type Impact LONG haul flights naturally have higher premium cabin allocation and attract customers with elite status due to ticket prices and flight duration SHORT haul flights are more democratized but still show loyalty patterns, especially on business routes C. Data-Driven Validation From the 10,000 flight dataset: Consistent patterns emerged: Tier percentages were remarkably stable within each group ($\pm 0.2\%$ variation) SHORT haul dominates: 59.75% of flights, requiring efficient lounge operations LONG haul consistency: 40.25% of flights with ~290 average seat capacity D. Scalability & Flexibility No dependency on specific aircraft types (which may change with fleet modernization) No dependency on specific destinations (which may be added/removed from network) Simple to apply: Only need departure time and route distance - both always known

3. What assumptions did you make about passenger eligibility ?	Core Assumptions: A. Historical Data Represents True Demand The existing TIER1/2/3_ELIGIBLE_PAX columns in the dataset reflect actual lounge eligibility based on: Ticket class (First, Business, Premium Economy, Economy) British Airways Executive Club tier (Blue, Bronze, Silver, Gold, Gold Guest List) Partner airline status (oneworld Emerald, Sapphire, Ruby) B. Tier Distribution Assumptions Tier 1 (Concorde Room) - Average 1.5-1.7%: First Class passengers Gold Guest List members oneworld Emerald on premium tickets Note: Currently hypothetical at Terminal 3 Tier 2 (First Lounge) - Average 20.3-20.5%: Business Class passengers Gold Executive Club members oneworld Emerald members Silver members on long-haul business class Tier 3 (Club Lounge) - Average 77.9-78.1%: Business Class passengers (overlap with Tier 2) Club World (Business) passengers Silver and Bronze Executive Club members oneworld Sapphire members Premium Economy on select routes C. Customer Behavior Assumptions Morning business travelers have slightly higher loyalty penetration Regular travelers accumulate status faster Corporate contracts often include premium cabin access Time of day affects passenger mix but not dramatically Evening/Sleeper flights show marginally higher Tier 1 (1.58% vs 1.47%) Suggests premium travelers prefer overnight comfort
--	--

4. How can your model apply to future or changing flight schedules?	Application Framework: Step 1: Classify Any Future Flight For any new or changed flight, you only need two pieces of information: Is it SHORT or LONG haul? (based on distance/destination) What time does it depart? (assigns PB, OP, or ES category) Step 2: Apply Lookup Table Percentages Match the flight to one of the 6 groups Apply the corresponding Tier 1%, Tier 2%, Tier 3% to estimated passenger count Calculate lounge demand Step 3: Aggregate Across Schedule Sum all individual flight estimates by tier This gives total daily/weekly/monthly lounge demand Practical Use Cases: Scenario A: Network Expansion "BA adds 5 new flights to Tokyo (LONG haul), all departing 09:00-10:00" Classification: LONG - Peak Business Apply: 1.48% Tier 1, 20.42% Tier 2, 78.10% Tier 3 If each flight averages 250 passengers: 3.7 Tier 1, 51 Tier 2, 195 Tier 3 per flight
--	---